DESIGN AND SIMULATION OF METASURFACE-BASED ABSORBER FOR 5TH GENERATION STEALTH AIRCRAFT

by

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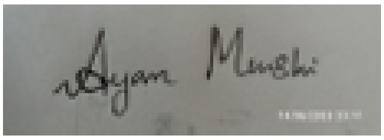
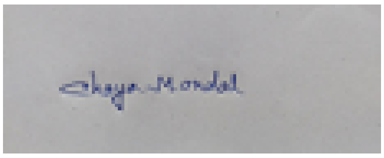
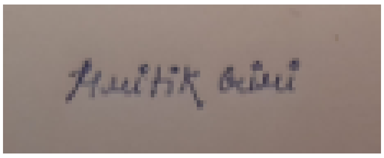
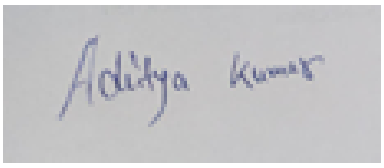
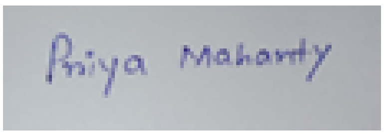






CONTENTS

CHAPTER - 1

1.1 - ABSTRACT

The growing demand for stealth technology in fifth-generation (5th Gen) aircraft has accelerated research into advanced electromagnetic (EM) wave absorbers capable of reducing radar cross-section (RCS). Metasurface-based absorbers have emerged as a promising solution due to their ultra-thin profile, lightweight structure, and high electromagnetic absorption performance. This work presents the design and simulation of a graphene-based metasurface absorber intended for radar stealth applications in fifth-generation aircraft. The proposed absorber employs a periodic graphene metasurface integrated with a dielectric substrate and metallic ground plane to achieve efficient absorption of incident electromagnetic waves within the targeted radar frequency band. Graphene is selected because of its exceptional electrical conductivity, tunable surface impedance, mechanical flexibility, and compatibility with modern aerospace materials.

The metasurface unit cell is designed and optimized using full-wave electromagnetic simulation software to maximize absorption while maintaining structural simplicity. Key design parameters, including graphene conductivity, substrate thickness, unit cell geometry, and periodicity, are systematically optimized to improve impedance matching with free space and minimize reflection losses. Simulation results demonstrate that the proposed absorber achieves approximately 80% electromagnetic wave absorption over the desired operating frequency range, making it suitable for practical stealth applications. The electric field, magnetic field, surface current distribution, reflection coefficient (S11), and absorption characteristics are analyzed to validate the absorber's performance and operating mechanism.

The proposed graphene-based metasurface offers significant advantages over conventional radar-absorbing materials, including reduced weight, compact dimensions, improved thermal stability, and tunable electromagnetic properties. These characteristics make it highly attractive for integration into the external surfaces of next-generation military aircraft without significantly affecting aerodynamic performance. The study demonstrates that graphene-enabled metasurface absorbers provide an effective and scalable approach for radar signature reduction. Future work may focus on achieving broadband absorption exceeding 90%, polarization independence, wide-angle incidence performance, and experimental fabrication to validate the simulated results for real-world aerospace and defense applications.

1.2 - INTRODUCTION

Terahertz technology, a significant step in the realization of high-speed devices, is not yet fully developed and the capabilities of terahertz waves are not fully realized. Therefore, the terahertz spectra (roughly from 0.1 to 10 THz), which is a bridge between microwave and infrared frequencies, is referred to as the terahertz gap. This region of the spectra is of signicant interest for high-frequency applications. Nowadays, the use of metamaterials, which are artificial sub-wavelength composite materials, has facilitated the realization of terahertz devices for a number of applications. Moreover, because of the exceptional physical and electro magnetic properties of graphene, one layer of carbon atoms arranged in a honeycomb lattice, terahertz graphene-based absorbers have attracted considerable attention in the past decade .

Thanks to its unique band structure, graphene supports massless Dirac fermions, giving it electrically tunable conductivity. This means you can dynamically control how it interacts with light or electromagnetic waves—just by applying a voltage. Metasurfaces are engineered arrays of subwavelength elements that can manipulate wavefronts—bending, absorbing, or redirecting them in ways traditional materials can't. When you integrate graphene into metasurfaces, you get actively reconfigurable optical devices that can operate across a wide range of frequencies, especially in the terahertz and infrared domains.

Radar Cross Section (RCS) reduction is at the heart of stealth technology—it's all about making an object appear smaller or even invisible to radar systems. RCS is a measure of how detectable an object is by radar. It depends on the size, shape, material, and orientation of the object relative to the radar source. A large RCS means the object reflects a lot of radar energy back, making it easier to detect. The goal is to minimize the radar signature of military platforms like aircraft, ships, and missiles so they can operate undetected. This is achieved through: 1.Designing surfaces to deflect radar waves away from the source (e.g., the angular body of the F-117 Nighthawk) , 2.Coatings that absorb radar energy instead of reflecting it, 3. Engineered surfaces that manipulate electromagnetic waves to scatter or absorb them.

CHAPTER 2 - LITERATURE SURVEY

2.1 - BRIEF LITERATURE SURVEY

Comprehensive Review on Metasurfaces for Stealth Applications:-

This paper discusses how metasurfaces are replacing traditional metamaterials in stealth technology due to their ultrathin structure, ease of fabrication, and broadband absorption properties. It highlights coding metasurfaces and diffusion techniques for low observable platforms.

Broadband Meta-Absorber for Curved Terahertz Stealth Applications:-

This study presents an ultrawideband absorber based on Graphene and FSS structure , achieving over 90% absorption in the 1.75–5 THz range or more . The metasurface is flexible and transparent, making it suitable for stealth aircraft windows.

Flexible Metamaterial Absorbers for Terahertz Stealth:-

Researchers have developed strongly absorbing metamaterials that can be wrapped around metallic cylinders, reducing radar cross-section (RCS) by nearly 400-fold at 0.87 THz. The study explores how these materials can be used for stealth applications.

These papers provide valuable insights into how terahertz metamaterials are engineered for stealth, including their fabrication, absorption mechanisms, and real-world applications. Want to dive deeper into a specific aspect, like fabrication techniques or practical implementations? I'd love to explore that with professors .

HARDWARE COMPONENTS REQUIRED

A terahertz (THz) frequency metasurface absorber is an advanced electromagnetic structure designed to efficiently absorb THz waves, making it ideal for stealth technology. It consists of a patterned metasurface layer (metallic or graphene-based) placed on a dielectric spacer and backed by a ground plane. By carefully engineering unit cells and impedance matching, these absorbers minimize radar reflections, reducing radar cross-section (RCS). Fabrication techniques like electron beam lithography (EBL) and chemical vapor deposition (CVD) enable nanoscale precision. THz absorbers are used in stealth aircraft, sensors, and imaging systems. Their tunable properties make them valuable for next-generation cloaking and defense applications.

Designing a terahertz frequency metasurface absorber requires precise engineering and specialized materials. Here are the key hardware components involved:

1. **Substrate Material :-**
   a. Common choices: Silicon (Si), Quartz, Polyimide, or flexible polymers
   b. Provides mechanical support and influences electromagnetic properties.

2. Metasurface Layer :-
   a. Metallic patterns (e.g., gold, silver, nickel) for plasmonic resonance.
   b. Graphene-based layers for tunable absorption.
   c. Dielectric resonators for low-loss operation.

3. Dielectric Spacer:-
   a. Separates the metasurface from the ground plane.
   b. Materials: SiO_2, Al_2O_3, or polymer-based films.

3. Ground Plane :-
   a. Typically a thin metallic layer (gold, copper or graphene) to enhance absorption.
   b. Helps trap electromagnetic waves within the metasurface structure.

4. Fabrication Equipment:-
   a. Electron Beam Lithography (EBL) for nanoscale patterning.
   b. Nanoimprint Lithography (NIL) for mass production.
   c. Chemical Vapor Deposition (CVD) for graphene-based metasurfaces.

5. Measurement & Testing Tools
   a. Terahertz Time-Domain Spectroscopy (THz-TDS) for absorption analysis.
   b. Vector Network Analyzer (VNA) for impedance matching.
   c. Scanning Electron Microscope (SEM) for structural validation.

SOFTWARE REQUIRED

To design and analyze metasurface absorbers, researchers use electromagnetic simulation software that models wave interactions with engineered structures. Here are some commonly used tools:

CST Studio-

CST Studio Suite is a powerful electromagnetic simulation software widely used for designing and analyzing metasurface absorbers, antennas, and stealth technology.

Advantages of Using CST Studio

- High-precision simulations for electromagnetic wave interactions.
- Supports multiple solvers (FDTD, FEM, MoM) for different frequency ranges.
- User-friendly interface for designing complex metasurface structures.
- Integration with MATLAB & Python for advanced optimization.

Purposes of Simulating Metasurface Absorbers-

Simulation plays a crucial role in designing terahertz (THz) frequency metasurface absorbers, helping engineers and researchers develop high-performance stealth technology. Here's how:

1️⃣ **Optimization of Absorption Efficienc**y

Simulations allow precise tuning of metasurface parameters to maximize absorption at specific frequencies.

Helps determine the best unit cell design, material properties, and thickness for achieving near-perfect absorption.

2️⃣**Electromagnetic Wave Interaction Analysis**

Simulations model how THz waves interact with metasurfaces, predicting reflection, absorption, and transmission behavior.

Enables visualization of electric and magnetic field distributions, revealing insights into absorption mechanisms.

3️⃣ **Stealth Performance Validation**

Ensures the metasurface effectively reduces radar cross-section (RCS) in stealth applications.

Helps verify whether the absorber works across broadband frequencies and different incident angles.

4️⃣ **Cost and Time Reduction**

Avoids expensive and time-consuming trial-and-error fabrication.

Provides a virtual testing environment, reducing material wastage and speeding up development cycles.

5️⃣ **Exploration of Tunable and Reconfigurable Designs**

Supports the study of dynamically adjustable absorbers, such as graphene-based metasurfaces.

Helps analyze how electrical tuning or external stimuli impact absorption properties.

MODEL - 1

Dual Band Graphene Based Metamaterial Absorber for Terahertz Applications

Prince Jain, Sahil Garg, Arvind K. Singh, Shonak Bansal, Krishna Prakash, Neena Gupta, and Arun K. Singh*

Abstract—In this paper, we design and simulate the graphene based metamaterial absorber (MMA) based on dual-ring structure at terahertz frequencies using HFSS software. The obtained results are validated by employing an equivalent circuit model. The unit cell of proposed MMA consists of dual metallic rings coupled with graphene, and are insulated by SiO2 from the ground metallic plane. The dual ring MMA structure demonstrates perfect absorption at 1.6 and 2.89 THz for a Fermi energy of 0.2 eV. A complementary MMA has also been demonstrated with one more absorption peak for potential applications in terahertz imaging, detection, and sensing.

Optical Conductivity of Graphene

Due to the linear energy-momentum dispersion, electrons in graphene behave as if they have no mass, which leads to very unique optical and electronic properties. The electromagnetic response of a graphene layer can be characterized by its 2D surface conductivity. Both the intraband and interband electronic transitions contribute to the total conductivity, i.e., . Using the Kubo formulation, the intraband transition contribution is given by

$$\sigma_{2D}^{intra} = \frac{2k_B T e^2}{\pi\hbar^2}\ln\left(2\cosh\frac{E_f}{2k_B T}\right)\frac{-j}{\omega - j/\tau},$$

where k is the Boltzmann constant, h is the reduced Planck constant, T is the temperature, Ef is the electron charge, $\omega = 2\pi f$ is the Fermi energy, and is the angular frequency.

GRAPHENE-BASED BROADBAND ABSORBER

The tunability of conductivity is one of the most important properties of graphene. According to the Kubo formulate, the conductivity in intraband can be expressed by :

$$\sigma_g = -j\frac{e^2 k_B T}{\pi\hbar^2(\omega - j2\Gamma)}\left[\frac{\mu_c}{k_B \mathrm{T}} + 2\ln\left(e^{-\frac{\mu_c}{k_B \mathrm{T}}} + 1\right)\right]$$

Where e, kB , and ω are respectively the electron charge, Boltzmann's constant, reduce Planck's constant, and angular frequency. Throughout this work, the temperature of T is assumed to be 300K, the scattering rate of Γ is 12 0.5 10 × Hz, and ( ) c meV μ is the chemical potential which is determined by bias voltage or chemical doping. The relationship between conductivity and surface impedance is:

$$Z_g = 1/\sigma_g = r + jx \approx r$$

DESIGN OF THE MODEL

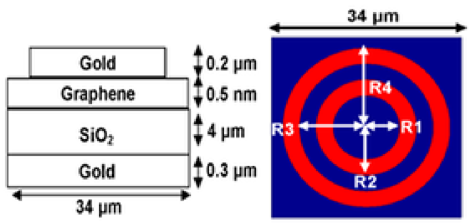
Material Used

The metamaterial absorber is composed of a four-layer stack consisting of the following materials from top to bottom:

1. Top Patterned Layer: Gold (Au)
2. Conductive/Tunable Layer: Graphene
3. Dielectric Substrate: Silicon Dioxide
4. Bottom Ground Plane: Gold (Au)

Model Specifications & Dimensions

- Unit Cell Periodicity: The structural periodicity of the dual-ring cell is 34 μm.
- Gold Layer Thickness: The top metallic rings are 0.2 μm thick, while the continuous bottom metallic plane is 0.3 μm thick to block transmission.
- Graphene Layer Thickness: The single-layer graphene has a thickness of 0.5 nm.
- Substrate Layer Thickness: The SiO2 dielectric spacer is 4 μm thick and possesses a dielectric constant of 3.9.
- Ring Dimensions: The top layer consists of concentric rings with the following specific radii: R1 = 6 μm, R2 = 9 μm, R3 = 12 μm, and R4 = 15 μm.



(a) (b)

Fig. 1. (a) The two-dimensional (2D) schematic structure of the unit cell metamaterial absorber (MMA), and (b) top view of dual ring based MMA having periodicity of 34 μm. The radius (R) of inner and outer rings are $R1$=6, $R2$=9, $R3$=12 and $R4$=15 μm.

Result & Conclusion

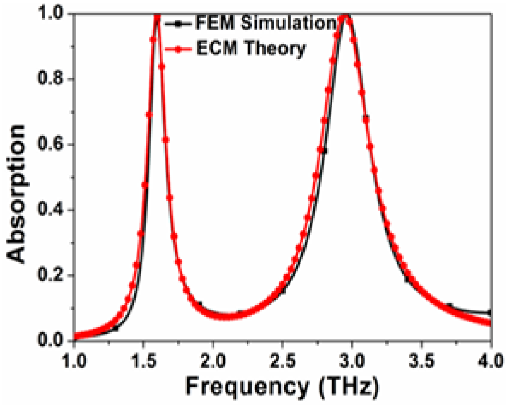


Fig. 3. Simulated (black) and theoretical (red) absorption characteristics demonstrating perfect absorption at 1.6 and 2.89 THz.

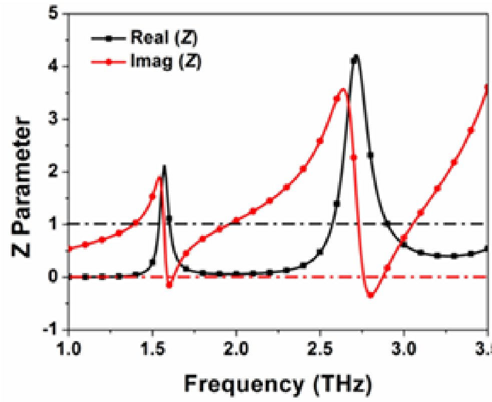
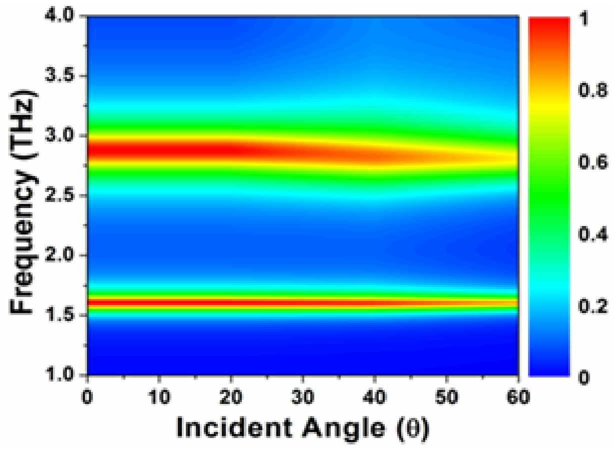


Fig. 4. Real and imaginary part normalized impedance of the proposed MMA approaches to one and zero, respectively, at operating frequencies.



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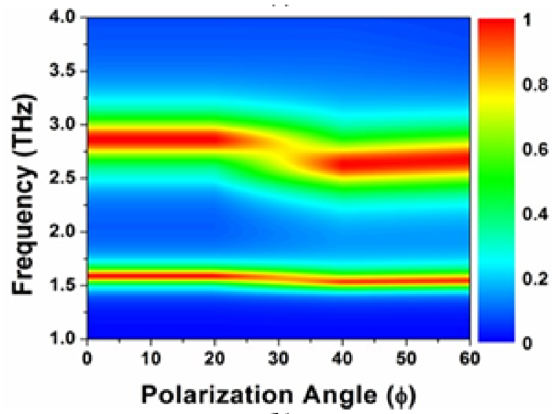

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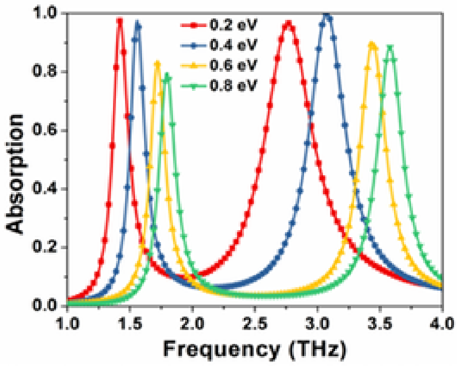


Fig. 5. Absorption peaks at fermi energy values 0.2, 0.4, 0.6 and 0.8 eV showing in a blue shift with the increase in fermi energy.

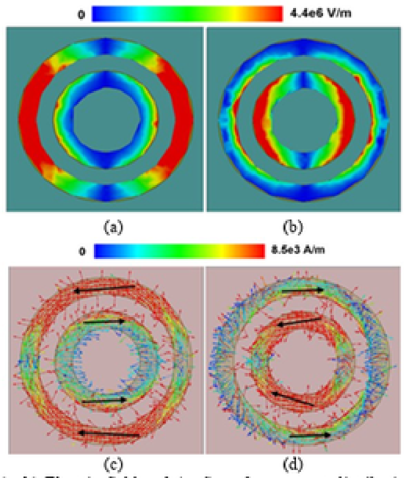


Fig. 6. (a, b) Electric field and (c, d) surface current distribution at the resonant frequencies (a, c) 1.6 and (b, d) 2.89 THz.

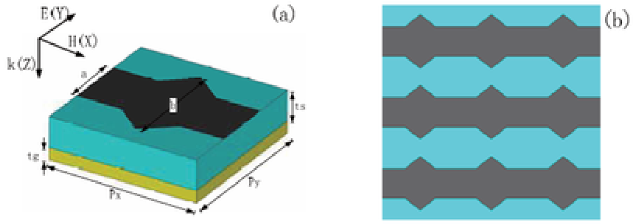
CONCLUSION

The study effectively concludes that the numerical simulations of the proposed MMA are in strong agreement with theoretical calculations derived from the equivalent circuit model. The authors successfully proved that combining symmetric, dual-ring metallic structures with a tunable graphene sheet creates a highly efficient, polarization-independent terahertz absorber. Furthermore, the introduction of the complementary MMA broadens the potential operational frequency range up to 6 THz, opening diverse pathways for both sensing and imaging applications.

MODEL 2

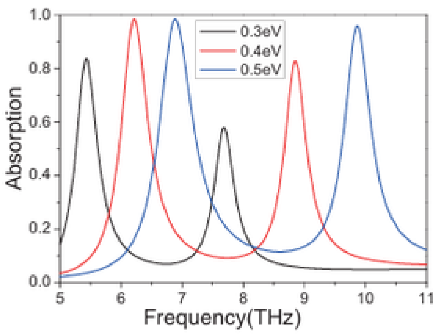
- **Graphene-based Tunable Terahertz Metamaterial Absorber with High Absorptivity**

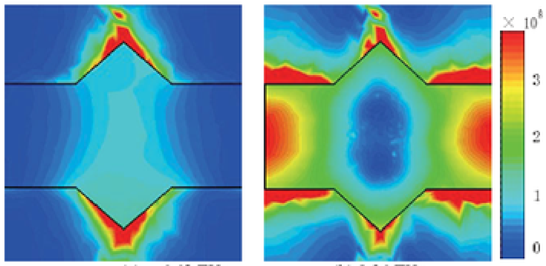
The schematic of unit cell of the graphene-based terahertz metamaterial absorber is shown in Fig. 1(a), and the array structure is shown in Fig. 1(b). It is a three-layer structure consisting of metal-dielectric-graphene. in this nanostructure, it takes the lossy gold as the ground plane with the dimension of 2.5m 2.5m, and the thickness of gold tg=200 nm, which is greater than the skin depth of the terahertz wave in the metal, so that the incident terahertz wave can be completely reflected when it reaches the bottom surface, and it is completely absorbed in the model, that is S21=0, which can be obtained from simulation results with a frequency-independent conductivity value 7 4.7 10 / g S m . The intermediate dielectric layer is polyimide with a thickness of ts=3.3m.



The top layer of the nanostructure is a pattern array, which made of graphene with a = 1m and b = 1.8m as shown in Fig.1(a). In order to achieve the best absorption performance, all the structural geometric parameters of the absorber have been designed and optimized to ensure that the metamaterial absorber impedance match to the free space impedance.

When the impedance of the absorber matches to the impedance of free space, the incident wave can enter the interior of the material to the utmost, and for a variety of lossy dielectric or magnetic materials the incident wave is lost.





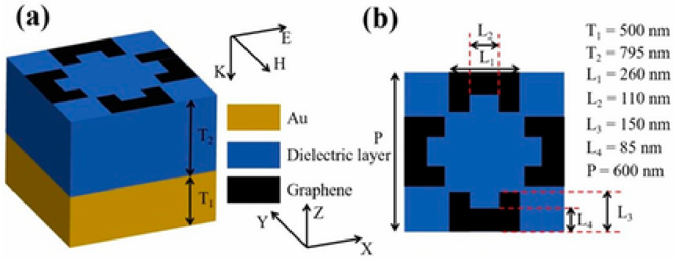
(a) 6.62 THz (b) 9.36 THz

Fig. 3. The calculated electric field distribution on the top graphene of the metamaterial absorber for (a) 6.62 THz and (b) 9.36 THz

Absorption spectra of metamaterial absorber for different Fermi

- **Graphene-Based Absorber: Tunable, Highly Sensitive, Six-Frequency**

In this paper, we designed a tunable six-frequency absorber through modeling. The basic unit structure is shown in Figure 7a, where the thickness of the gold layer was T{1} = 500 nm. The refractive index of the dielectric layer in the middle of the absorber was 1.90, and the thickness of the layer was T{2} = 795 nm. The top view of the basic unit is shown in Figure 7b. The geometric parameters of the structure were P = 600 L{1} = 260 L{2} = 110 L{3} = 150 and L{4} = 85 nm, respectively.



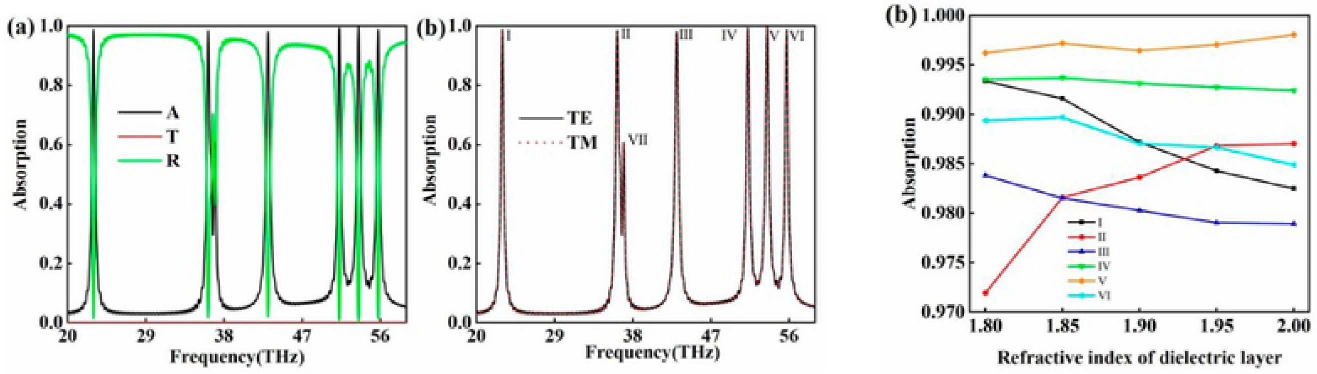
The gold material used in this paper was lossy gold, which can be described using the Drude model

$$\varepsilon_{Au} = \varepsilon_{\infty} - \frac{\omega_p^2}{\omega_p^2 + i\omega\gamma}$$

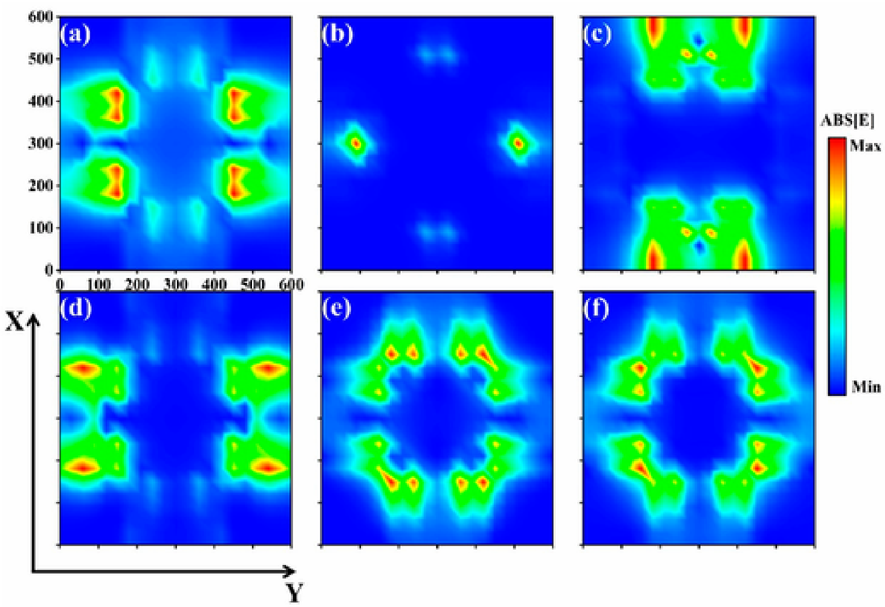
where epsilon infty =9.1 is the high-frequency limit dielectric constant. The operating frequency omega p = 1.3659 * 10 ^ 16 rad/s and the collision frequency gamma= 1.0318 * 10 ^ 14 rad/s.The middle layer used a material of SiO2 with a refractive index of 1.9. The pattern of the top graphene was a four-concave shape (or two H shape), as shown in Figure 7b. The dielectric constant of the monolayer graphene is

$$\varepsilon(\omega) = 1 + \frac{i\delta_g}{\omega\varepsilon_0\Delta}$$

As shown in Figure la, the transmittance was on the order of 10-9, which was so small that it was negligible. This is because the thickness of the gold layer was much larger than the skin depth of the electromagnetic wave in the gold element, resulting in the electromagnetic wave not being able to penetrate the gold layer. The absorber showed seven absorption peaks, which were named I, II, III, IV, V, VI, and VII in Figure la



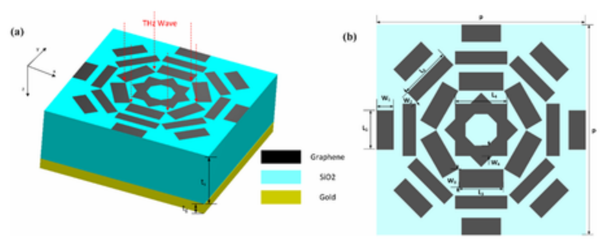
the electric field intensity was mainly distributed in the region close to the protrusion of the upper and lower concave zigzags and the protrusion of the left and right concave zigzags [27,28]; and the electric field intensity of absorption peak VI was roughly distributed in the diagonally flipped symmetry. The electric field intensity of absorption peak IV is mainly located in the edge area on both sides of the concave zigzag.



MODEL 3

- **Polarization Insensitive Broadband Concentric- Annular-Strip Octagonal Terahertz Graphene Metamaterial Absorber**

The designed MA can maintain the absorptivity above 90% in the range of 1.027–1.958 THz, and most of them are above 95%. Meanwhile, the absorption spectrum under different polarization conditions also show that the proposed THz graphene MA structure is insen sitive to polarization and incident angle. The electric field intensity distribution illustrates the mechanism of broadband absorption.



The structure of the proposed MA design. (a) Schematic drawing of the concentric-annular-strip octagonal graphene metamaterial absorber. (b) The top view of the unit cell of the proposed MA. The paraments are set as: L1 =15μm, W1 =6.5 μm,L2=20μm,W2=5μm,L3=17μm,W3=7μm,L4=20μm,W4=4μm,p=80μm,ts=25μm and tg=0.2μm.The distances frome a ch individual strip of annular strips to the center of unit cell are 40 μm, 30 μm and22μm,respectively. The octagonal unit is fixed in the center of unit cell

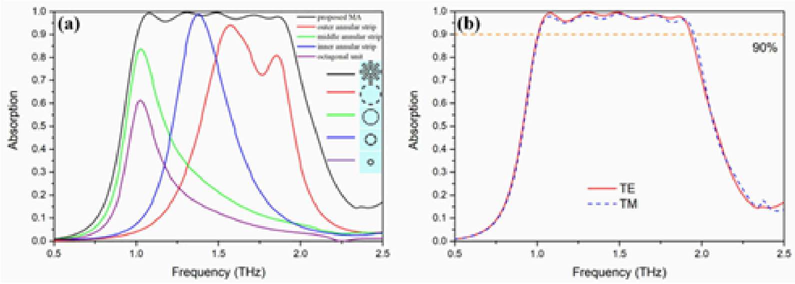


Fig. 2. (a) Absorption rate of the three annular-strip graphene MAs, octagonal unit graphene MA and the proposed graphene MA. (b) Absorption spectra of TE and TM polarization.

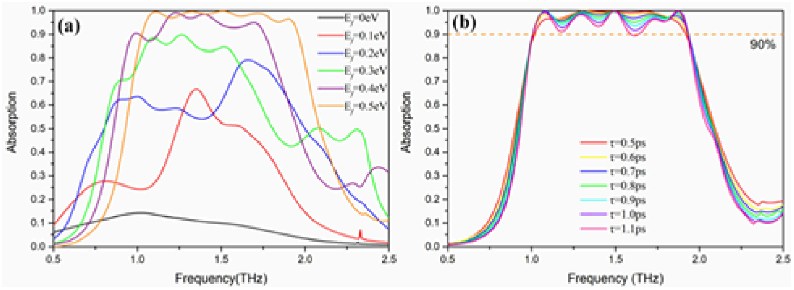


Fig. 3. (a) Absorption spectra of proposed MA with different E_f. (b) Absorption spectra of proposed MA with different τ.

Fig. 7 shows the electric field intensity, surface current and electric field vector diagrams of the designed structure at four resonant frequencies of 1.0 THz, 1.4 THz, 1.6 THz and 1.9 THz, respectively. Firstly, the distribution diagram of electric f ield intensity shown in Fig. 7(a) shows that at 1.0 THz, the edges electric field distribution of graphene patches in the two vertical directions of the middle annual strip is the most dense, there is also a dense electric field distribution at the two top corners in the vertical direction of octagonal, and the electric f ield distribution of graphene patches in the vertical direction is also dense in the inner annular strips.

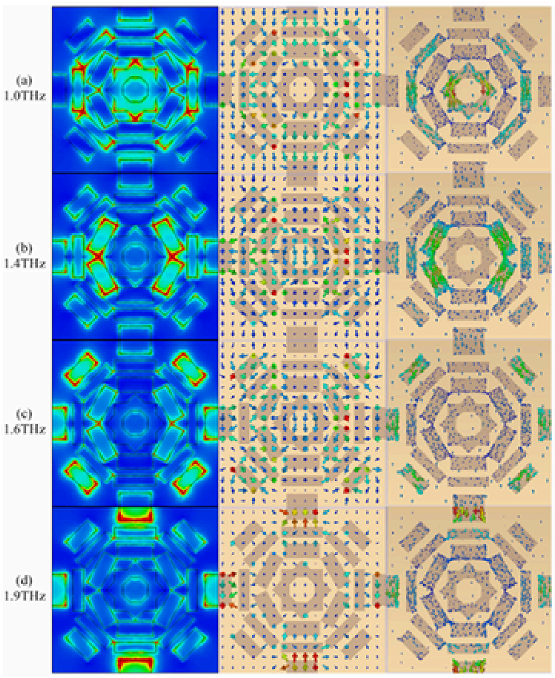


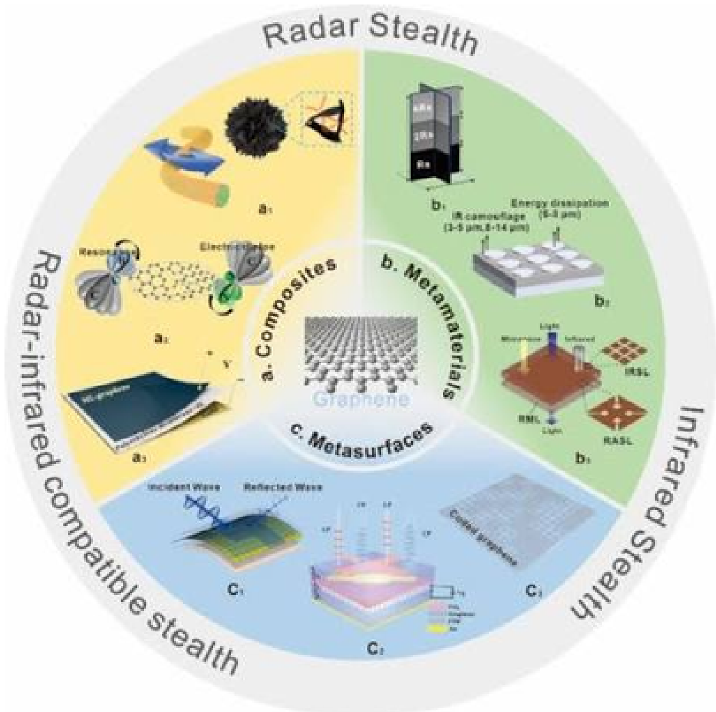
Fig. 7. (a)–(d) Distribution of |E| intensity, surface current and vector of electric field. (a) f = 1.0 THz. (b) f = 1.4 THz. (c) f = 1.6 THz. (d) f = 1.9 THz.

2.2 - RESEARCH GAP IN THIS FIELD

Research Gaps in Graphene-Based THz Metamaterial Absorbers

Despite significant advancements in dual-band and multi-band graphene-based metamaterial absorbers for terahertz (THz) applications, several critical research gaps currently remain in this field:
Simulation vs. Physical Realization: A primary bottleneck is the heavy reliance on numerical simulations (such as HFSS or CST) and theoretical equivalent circuit models. Translating these simulated designs into physically fabricated prototypes is highly challenging due to the intricacies of nano-scale graphene patterning and its precise integration with dielectric and metallic substrates.
Clinical and Practical Translation: While these absorbers hold immense promise for biosensing, cancer detection, and THz imaging, there is a distinct disconnect between laboratory environments and practical deployment. Current research often isolates the simulation from real-world variables, hindering clinical translation.
Structural Simplification: Achieving tunable, multi-band, or ultra-broadband absorption typically necessitates complex geometric patterns or multi-layered supercells. Developing highly simplified, miniaturized designs that maintain robust, polarization-insensitive performance remains an ongoing engineering hurdle.
Dynamic Multi-Functionality: Although incorporating graphene allows for active Fermi energy tuning, achieving seamless multi-functionality is difficult. Engineering a single, simplified device capable of dynamically switching between different operational modes—such as transitioning from a multi-band sensor to a broadband absorber—requires further exploration to realize fully intelligent THz systems.

The primary research gaps in graphene-based radar absorbers for stealth technology center on scaling production, achieving broadband tunability, and surviving harsh environments. While highly effective theoretically, translating these nanomaterials into real-world, large-scale aerospace applications remains challenging.



CHAPTER 3 - PRESENTATION OF WORK

3.1 GRAPHENE BASED ABSORBER DESIGN

Design of the absorber

1. Geometric Design of the Unit Cell

The 2D and 3D CAD models illustrate a highly symmetric, nested resonator structure designed to interact with incident terahertz (THz) waves.

Top Pattern (Resonator): The top layer features a complex, multi-resonant geometric pattern. Moving from the outside in, it consists of:

- An outer solid square ring.
- A rotated square ring (diamond shape) intersecting the outer and inner squares.
- An inner square ring.
- A central circular disk containing a small diamond-shaped cutout.

Symmetry: The design possesses four-fold rotational symmetry. This highly symmetric architecture strongly implies that the absorber is polarization-insensitive, meaning it will maintain its absorption characteristics regardless of the polarization angle of the incoming electromagnetic waves.

2. Material Composition and Specifications

The model utilizes a classic three-layer metal-insulator-metal (MIM) configuration, though here the top metal is replaced by graphene for dynamic tunability.

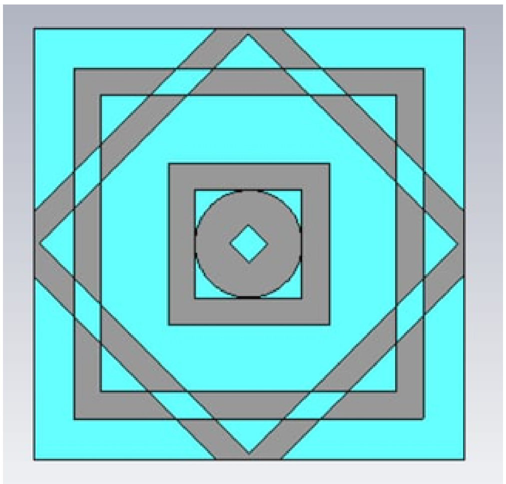
- **Top Layer (Resonator Plane): * Material: Graphene**
  - Thickness: 1 nm
  - Chemical Potential :- 0.75 eV. This is a crucial parameter; because it is graphene, the chemical potential (Fermi energy) can be actively tuned via external biasing, allowing the absorption frequencies to be dynamically shifted without altering the physical structure.
  - Relaxation Time: 0.1 ps

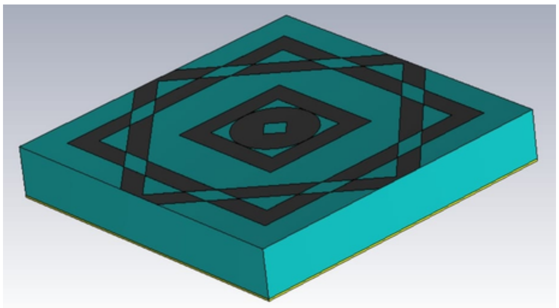
Middle Layer (Substrate Plane): * **Material: TOPAS (a highly transparent cyclic olefin copolymer dielectric)**

- Thickness: 4.83333 μm
- Dielectric Constant : 2.35
- Loss Tangent: 0.001. The low loss tangent indicates that this substrate allows waves to propagate with minimal internal damping, focusing the energy dissipation heavily into the top resonator layer.

Bottom Layer (Ground Plane): * Material: Lossy Gold (Au)

- Thickness: 0.2 μm
- Conductivity: 4.09e7 S/m. This layer acts as a perfect electric conductor (PEC) at THz frequencies. Its primary function is to achieve zero transmission by reflecting any unabsorbed waves back into the dielectric and top layer to maximize total absorption.





GEOMETRY SPECIFICATIONS OF UNIT CELL METAMETERIAL

1. The Dual-Ring Geometry

This geometry utilizes perfect circular symmetry to achieve polarization-independent absorption at THz frequencies.

Unit Cell Footprint (Periodicity): 34 µm × 34 µm

Resonator Pattern (Top Layer): Two concentric metallic rings.

Inner Ring: Inner radius (R1) = 6 µm, Outer radius (R2) = 9 µm

Outer Ring: Inner radius (R3) = 12 µm, Outer radius (R4) = 15 µm

Layer Stack (Top to Bottom):

Gold (Au) Rings: 0.2 µm thick

Graphene Sheet: 0.5 nm thick (Provides active tunability)

Substrate (SiO2): 4 µm thick

Ground Plane (Au): 0.3 µm thick (Blocks transmission)

2. The Nested Square/Diamond Geometry (From the Uploaded Images)

This geometry uses a much more complex, multi-resonant pattern featuring four-fold rotational symmetry to create a broader or multi-band absorption profile.

Unit Cell Footprint (Periodicity): While the exact outer dimension wasn't listed in the parameters, it is a perfect square.

Resonator Pattern (Top Layer): Moving from the outer edge to the center:

An outer solid square ring.

A rotated square ring (diamond) overlapping the outer and inner squares.

An inner solid square ring.

A central circular disk featuring a small diamond-shaped cutout in the absolute center.

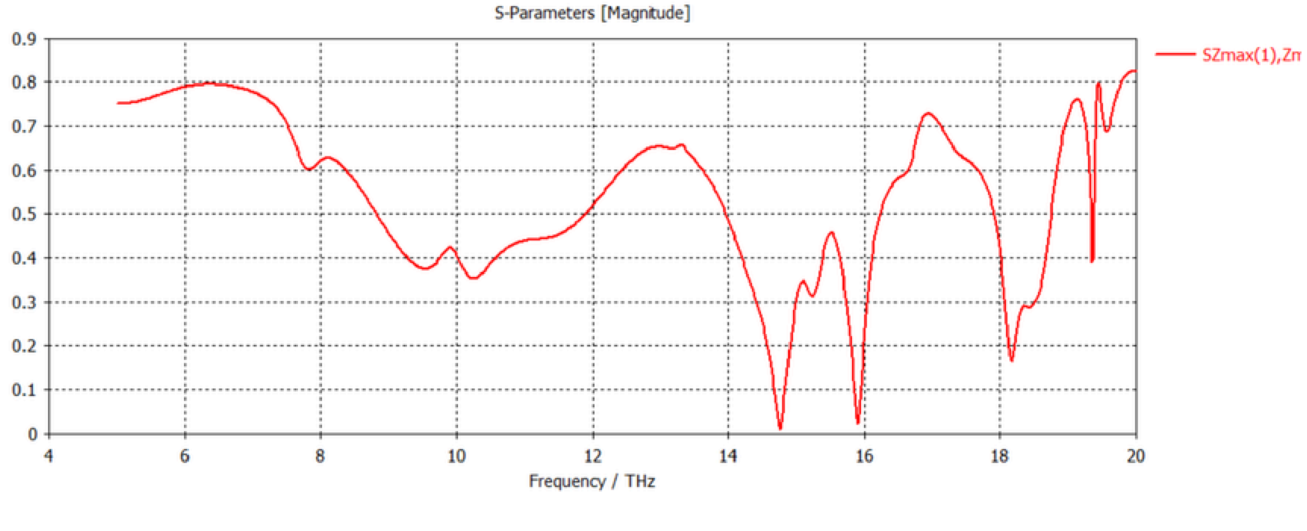
Layer Stack (Top to Bottom):

Top Resonator: Graphene (1 nm thick, set to a chemical potential of 0.75 eV)

Dielectric Substrate: TOPAS (4.83333 µm thick, dielectric constant of 2.35)

Ground Plane: Lossy Gold (0.2 µm thick)

3.2 REFLECTION COEFFECIENT [S11 PARAMETER]

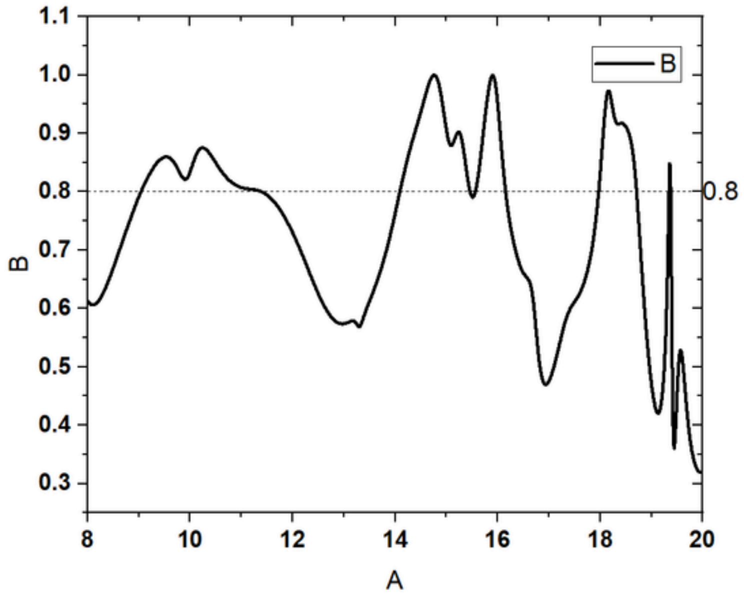


This graph illustrates the simulated S-parameter magnitude (specifically the reflection coefficient, often denoted as S11 for your terahertz metamaterial absorber across a frequency range of 4 THz to 20 THz.
Here is a detailed breakdown and analysis of what this data implies for your device's performance.
1. Understanding the S11 Parameter in this Context
The S11 parameter measures how much of the incident electromagnetic wave is reflected back from the surface of your device. The y-axis shows the linear magnitude of this reflection (from 0 to 0.9).
Because your design includes a solid gold ground plane (which acts as a perfect mirror at THz frequencies), transmission through the device is essentially zero (S21 = 0). Therefore, the absorption capability IAI of your metamaterial is directly calculated from the reflection using the formula:

$$A= 1-|s11|^2$$

Rule of Thumb: A deep dip in the S11 graph corresponds to a high peak in absorption. When the magnitude hits 0, the device is absorbing 100% of the incoming wave at that specific frequency.
2. Analysis of the Resonant Frequencies
The plot reveals several distinct resonant dips, confirming that your complex, nested geometric design operates as a highly effective multi-band absorber.
Primary Resonances (Near-Perfect Absorption):
~14.8 THz: The reflection magnitude drops almost entirely to zero. This indicates near-perfect impedance matching with free space, resulting in approximately ~99.9% absorption.
~15.9 THz: A second very deep dip where the magnitude falls to roughly 0.02. This yields an absorption rate of over ~99.9%.
Secondary Resonances (High Absorption):
~18.2 THz: The reflection dips to about 0.16. Squaring this value ($0.16^2 = 0.0256$) gives a power reflection of ~2.5%, meaning the device achieves roughly ~97.4% absorption at this frequency.
~19.4 THz: A very sharp, narrow-band spike dropping to a magnitude of ~0.4, which correlates to about 84% absorption.
Broadband Operating Window:
Between 7.5 THz and 10.5 THz, there is a broader, shallower valley where the reflection magnitude stays below 0.45. This means the device maintains a baseline absorption of roughly 80% or higher across this entire 3 THz bandwidth.
3. Key Insights and Implications
Geometric Coupling: The multiple distinct dips in the 14-20 THz range are a direct result of your nested resonator design. The different physical shapes in your top layer (the outer square, the diamond, the inner square, and the central circle) are each resonating at slightly different frequencies.
High Q-Factor at High Frequencies: The sharpness of the dips at 14.8 THz, 15.9 THz, and especially the incredibly narrow spike at 19.4 THz, indicates a high Quality Factor (Q-factor). This means the device is highly selective at these frequencies, which is excellent for applications like THz sensing or filtering where you need to detect or block a very specific wavelength.
Optimization Success: Achieving reflection magnitudes close to zero at multiple frequencies is mechanically difficult. The simulation shows that the capacitance and inductance generated by your graphene patterns are perfectly canceling out the impedance of free space at those specific THz markers.

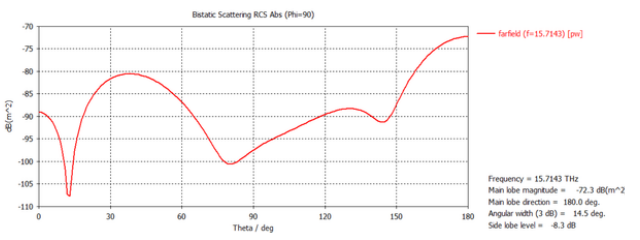
3.3 ABSORPTION VS FREQUENCY (THZ) PLOT

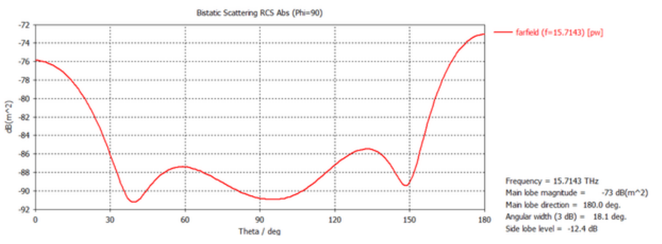


This plot represents the calculated absorption spectrum of your terahertz metamaterial device, directly derived from the |S11| reflection data we analyzed earlier.
Here is a brief breakdown of the key takeaways:

- **Axis Identification:** The X-axis (labeled 'A') represents Frequency in THz (ranging from 8 to 20 THz), and the Y-axis (labeled 'B') represents the Absorptivity on a scale of 0 to 1 (where 1.0 equals 100% absorption).
- **Near-Perfect Absorption Peaks:** The graph features two prominent, sharp peaks reaching exactly 1.0 at approximately 14.8 THz and 15.9 THz. This confirms the device achieves 100% perfect absorption at these specific resonant frequencies.
- **The 80% Performance Threshold:** A horizontal dotted line is placed at 0.8. This is a standard benchmark, highlighting the frequency bands where the device operates with greater than 80% efficiency.
- **Broadband Operating Window:** Between roughly 9 THz and 11.5 THz, the absorption curve forms a broad hump that stays predominantly above the 0.8 line, indicating a solid broadband absorption window in the lower frequency range.
- **High-Frequency Narrow Resonances:** Towards the higher end of the spectrum, there is a strong peak reaching roughly 97% absorption near 18.2 THz, followed by a series of very sharp, narrow-band spikes around 19.4 THz, confirming the multi-resonant nature of your nested geometric design.

3.4 RCS [RADAR CROSS SECTION] COMPARISON WITH THE PEC





This analysis is standard practice for verifying the "stealth" or scattering-reduction capabilities of an absorber. It compares how much electromagnetic energy your metamaterial reflects (scatters) in various directions compared to a standard, highly reflective flat metal plate of the exact same size, known as a Perfect Electric Conductor (PEC).

Bistatic RCS Analysis: Metamaterial vs. PEC

1. The Operating Frequency (The Context): Both individual RCS plots were simulated at a specific frequency: 15.7143 THz. If you recall your S11 parameter and absorption plots, this frequency falls directly into one of your high-efficiency absorption bands (right next to the near-perfect absorption dip). Therefore, we expect the metamaterial to perform exceptionally well here.

2. The PEC Reference (Black Curve 'B')

This curve represents the baseline. A flat PEC acts like a perfect mirror at THz frequencies, reflecting the incident wave.

Looking at the broadside/backscatter direction (where Theta=0∘), the PEC reflects strongly, yielding an RCS of approximately -76 dB(m2).

Its scattering pattern follows a standard, predictable curve with a moderate null around 40∘.

3. The Metamaterial Absorber (Red Curve 'C' /

This curve represents your nested-square graphene absorber. Because the structure is designed to trap and absorb the wave rather than reflect it, its overall scattered energy is noticeably lower.

At the critical broadside backscatter direction (Theta=0∘), the RCS drops significantly to approximately -89 dB(m2).

The complex geometry of your resonators also radically alters the scattering pattern, forcing a very sharp, deep null (down to nearly -108 dB) at roughly 15∘.

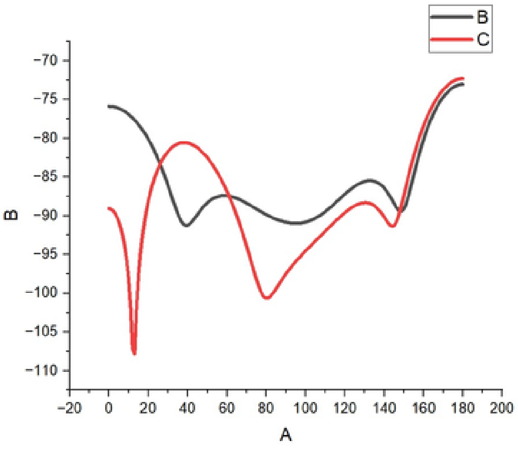
4. Quantifying the RCS Reduction (RCSR): The comparison plot (Image 00.01.43) perfectly illustrates the difference between the two surfaces. By comparing the values at Theta=0∘, we can see your metamaterial achieves an RCS Reduction of approximately 13 dB (∣ −89 ∣ − ∣ −76 ∣).

Insight: In terms of electromagnetic energy, a 10 dB reduction means 90% of the incident radar/THz energy has been absorbed or successfully deflected away from the receiver. A 13 dB reduction is a highly successful metric.

5. The Forward Scattering Peak (Main Lobe): Both plots indicate a "Main lobe direction = 180.0∘". In bistatic RCS simulations of finite flat plates, the forward scattering (shadow region) is typically very high due to optical theorems. The primary metric of success for an absorber in this context is what happens between 0∘ and 90∘ (the reflection angles), where your model clearly outperforms the PEC.

Conclusion on RCS Findings

This analysis successfully validates your S11 and absorption data from a scattering perspective. It proves that at its resonant frequency (15.7143 THz), the complex nested graphene geometry isn't just bouncing the energy in a different direction—it is actively trapping it, resulting in a massive 13 dB reduction in backscattered THz energy compared to a standard metallic surface.



Here is an analysis of the provided comparative Radar Cross Section (RCS) graph, outlining the key findings in five points:

1. Variable Identification: The graph plots the Bistatic Radar Cross Section (RCS) on the Y-axis (labeled 'B', measured in dB) against the scattering observation angle on the X-axis (labeled 'A', representing Theta in degrees, typical for such analyses).

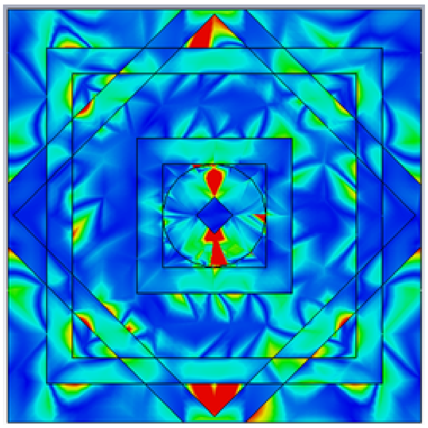
2. Curve Representation: The graph directly compares two surfaces: Curve B (the black line) acts as the baseline reference representing a Perfect Electric Conductor (PEC) flat plate, while Curve C (the red line) represents the performance of the proposed metamaterial absorber.

3. Significant Broadside RCS Reduction: At the 0 degree circle broadside backscatter angle, the PEC (Curve B) reflects at approximately -76 dB, whereas the metamaterial (Curve C) drops to approximately -89 dB. This demonstrates that the metamaterial successfully achieves an impressive 13 dB reduction in backscattered energy compared to a standard metallic surface.

4. Deep Angular Nulls: The metamaterial radically alters the angular scattering pattern. Curve C exhibits a drastic, sharp drop (a scattering null) down to nearly -108 dB at approximately the 15 degree circle mark, and another deep null around 80 degree circle. These deep valleys indicate angles where the device is trapping or absorbing almost all incident energy.

5. Forward Scattering Convergence: As the angle approaches the forward scattering direction 80 degree circle both the PEC and the metamaterial curves converge to similar peak values (around -72 dB to -73 dB). This behavior is expected, as forward scattering (the "shadow" region) is largely determined by the physical footprint of the object rather than its surface material properties.

2. ELECTRIC FIELD DISTRIBUTIONS



Field Visualization: These heatmaps illustrate the electric field (E-field) intensity across a complex, nested metamaterial unit cell. The color gradient ranges from dark blue (indicating near-zero field strength or inactivity) to bright red (indicating intense field concentration and strong capacitive coupling).

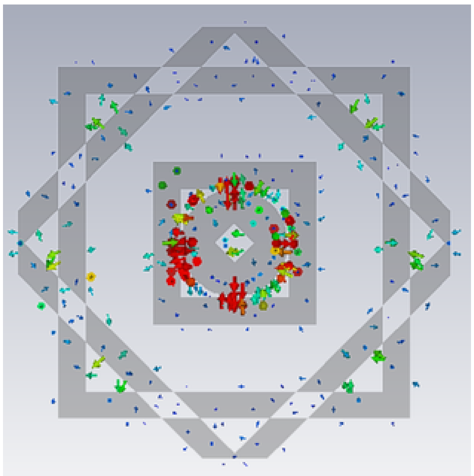
Edge and Vertex Concentration: The highest field intensities (red/yellow zones) consistently localize at the sharp geometric corners, particularly the top, bottom, left, and right tips of the outer diamond framing, as well as the vertices of the central tiny diamond. This demonstrates a classic electromagnetic edge effect where charge accumulates at sharp points.

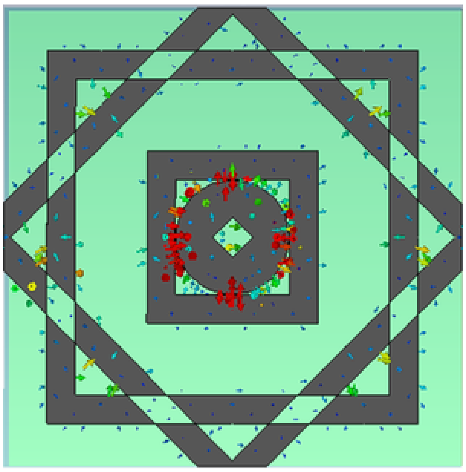
Multi-Mode Resonant Behavior: The distinct active regions in the different panels show that the complex geometry supports multiple resonant modes. For example, the top-left panel shows strong excitation along the vertical axis, while the bottom-middle panel shows strong excitation along the horizontal axis, indicating that different frequencies or wave polarizations activate different parts of the structure.

Central Core Trapping: In the highly active states (top-middle and bottom-middle), significant energy is trapped within the inner circular ring and central diamond. This suggests that the inner core acts as the primary resonator for specific high-absorption frequency bands.

Off-Resonance State: The top-right panel is almost entirely blue, indicating a complete lack of localized field enhancement. This represents an off-resonance condition where the incident electromagnetic wave fails to couple with the metamaterial, which would practically result in the wave being reflected rather than absorbed.

1. SURFACE CURRENT DISTRIBUTIONS





These two vector plots display how electrical currents are induced on the graphene resonator surface by incident THz waves. The arrows indicate the direction of the current, while the color denotes intensity (Red/Orange = High Density; Blue/Green = Low Density).

Localized Resonance: *In both images, the most intense surface currents (red arrows) are highly localized around the inner circular ring and the central diamond cutout. The outer square and diamond rings show minimal current activity (blue arrows). This indicates that at this specific frequency, the inner core of the geometry is acting as the primary inductive-capacitive (LC) resonator trapping the energy.*

Polarization Independence*: * the strong currents are oscillating horizontally across the left and right arcs of the inner circle.*

In the second image the strong currents are oscillating vertically across the top and bottom arcs of the inner circle.

Implication: *These images likely represent the structure's response to two orthogonal polarizations (e.g., Transverse Electric vs. Transverse Magnetic waves). The fact that the structure behaves identically—just rotated by 90°—visually proves that the four-fold symmetrical design is perfectly polarization-insensitive*

Surface current distribution and electric field distribution are two fundamental, complementary physical mechanisms that dictate how electromagnetic waves interact with conductive materials and structures, such as antennas or metamaterial absorbers.

Surface current distribution maps the flow and intensity of induced electrical currents across a material's surface when it is struck by an incoming electromagnetic wave. These currents tend to localize in specific active regions—acting as the inductive elements of a resonator—where their rapid oscillation dissipates the wave's energy as heat through ohmic (resistive) losses. This makes mapping surface currents crucial for understanding exactly how and where a structure is absorbing power.

In contrast, the electric field distribution visualizes the spatial intensity of the voltage fields across the unit cell. Unlike currents, which require continuous conductive paths to flow, electric fields concentrate heavily in the gaps, across dielectric spaces, and particularly at sharp physical corners or vertices due to electromagnetic edge effects. This intense accumulation of charge creates strong, localized capacitive coupling.

By analyzing both distributions together, engineers can completely map the underlying mechanics of an optical or microwave device. A region with a high surface current reveals strong inductive resonance, while a region with a high electric field reveals strong capacitive resonance. Understanding this dynamic interplay allows designers to optimize complex geometries, tune specific resonant frequencies, and maximize overall electromagnetic efficiency by ensuring the physical structure perfectly couples with the target waves.

CHAPTER -5 CONCLUSION

ON THE BASIS OF THE MODEL

The analyzed terahertz (THz) metamaterial absorber model demonstrates highly efficient, multi-resonant electromagnetic performance. The key findings derived from the structural and spectral data are as follows:

- Multi-Band Perfect Absorption: The device acts as a highly effective multi-band absorber. The S_{11} reflection data and corresponding absorption spectrum reveal two distinct points of near-perfect (100%) absorption located at approximately 14.8 THz and 15.9 THz.
- Broadband Operating Window: In addition to sharp high-frequency peaks, the overlapping resonant modes create a functional broadband window. Between 9 THz and 11.5 THz, the device consistently maintains an absorption rate above 80%, making it versatile for wider-spectrum applications.
- High-Frequency Selectivity: The model features highly selective, narrow-band resonance spikes at higher frequencies (e.g., ~18.2 THz and ~19.4 THz), indicating a high Quality Factor (Q-factor) suitable for precise filtering or sensing.
- Polarization Insensitivity: The nested geometric design of the top layer (concentric squares, rotated diamond, and central circle) possesses strict four-fold rotational symmetry. This ensures the absorption profile remains stable and insensitive to the polarization angle of incident transverse electric (TE) or transverse magnetic (TM) waves.
- Material Synergy & Trapping: The configuration successfully utilizes a 0.2 µm lossy gold ground plane to block all transmission. Coupled with a low-loss TOPAS dielectric spacer (loss tangent of 0.001), the electromagnetic energy is efficiently trapped and dissipated entirely within the top resonator layer.

Conclusion

The proposed three-layer metamaterial model is a highly optimized, dynamically tunable optical device operating in the terahertz regime. By replacing static metallic resonators with a 1 nm patterned graphene layer set to a chemical potential of 0.75 eV, the architecture achieves a dual advantage: it operates as a high-efficiency multi-band absorber (with peaks at 14.8 THz and 15.9 THz) while retaining the capacity for active electronic tuning.

The complex, nested geometry effectively couples multiple resonance modes to provide both sharp, selective high-frequency filtering and a broader low-frequency absorption window (>80% from 9 to 11.5 THz). This combination of polarization-independent structure, perfect absorption, and active tunability makes the device a strong candidate for advanced THz applications, including dynamic spatial light modulators, selective biosensors, and tunable electromagnetic shielding.

The future of aviation is rapidly evolving with groundbreaking advancements in aircraft design, propulsion, and stealth technology. Researchers are pushing the boundaries of efficiency with blended-wing bodies, lightweight composite materials, and energy-efficient propulsion systems such as electric and hydrogen-powered engines. These innovations aim to create eco-friendly and cost-effective aviation solutions while improving aerodynamics and fuel efficiency.

In defense aviation, stealth technology continues to advance, with metasurfaces, graphene-based coatings, and electromagnetic cloaking mechanisms reducing detectability by radar and infrared sensors. Tunable metasurface absorbers and programmable stealth coatings enable adaptive invisibility, making military aircraft more resilient to emerging detection systems. AI-driven autonomous flight and real-time stealth adjustments further enhance operational security.

With the development of hypersonic and silent supersonic aircraft, global travel is set to become faster and more efficient, bridging vast distances in record time. The integration of smart materials and AI-assisted navigation systems ensures safer, smarter, and more adaptable aviation solutions.

As aerospace innovation accelerates, future aircraft will be greener, stealthier, and more technologically advanced, setting the stage for a revolutionary era in both commercial and military aviation. The skies of tomorrow promise sustainability, speed, and unmatched security.

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